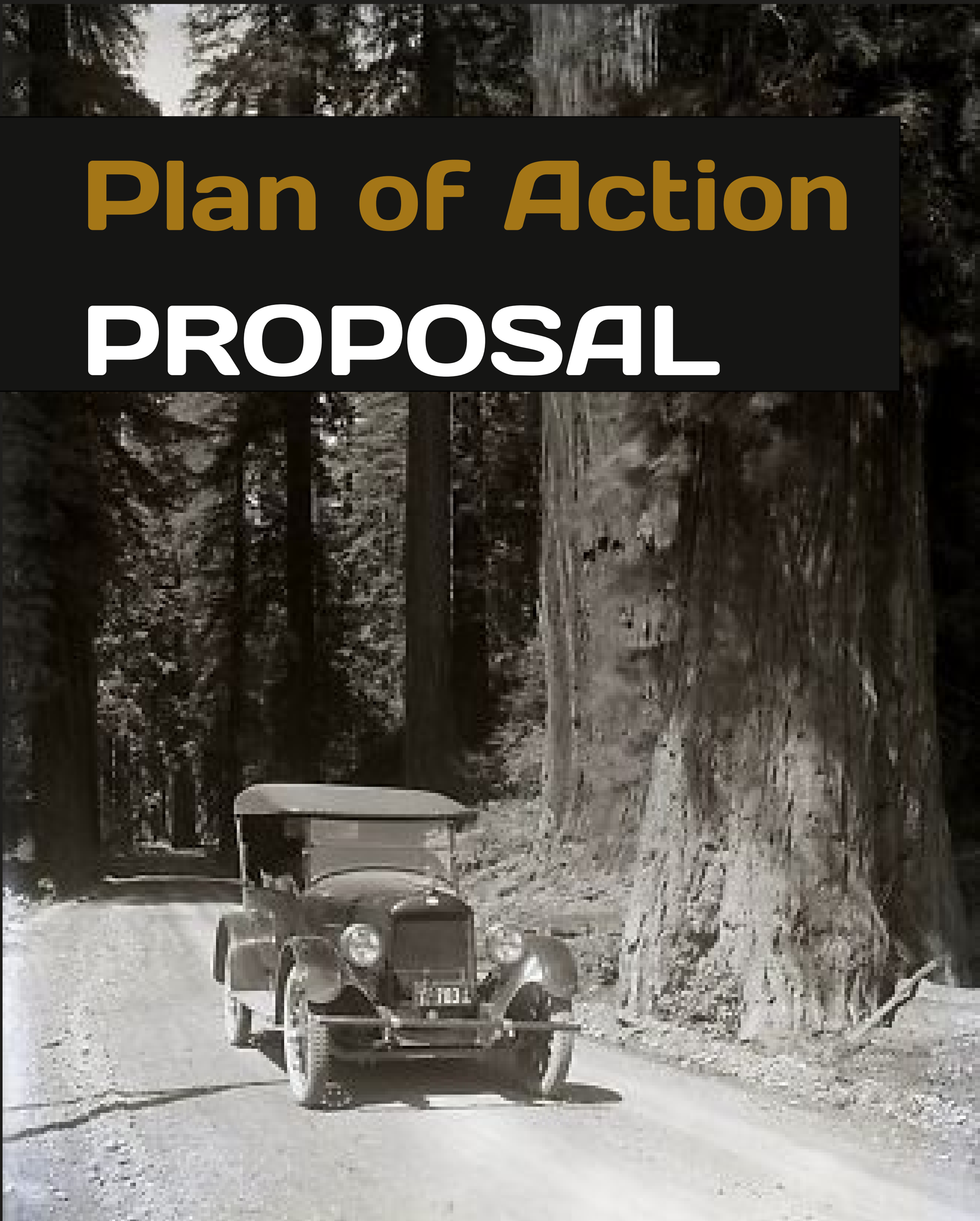
NSCC

Data Science in Business

Plan of Action

PROPOSAL



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Executive Summary

Timeless Transport Models has been proudly serving customers on a Business to Business basis. This document outlines the strategic plan to grow profits between June 2005 and December 2005. Drawing in four integrations of analyses; resource allocation, sales performance, and forecasting. Identifying data-driven recommendations designed to maximize revenue, reduce costs, and capitalize on seasonal demand patterns.

Business Context

Timeless Transport Models is a B2B manufacturer and distributor of modeled vehicles operating three factories across six product lines: Classic Cars, Vintage Cars, Motorcycles, Planes, Ships, Trucks & Buses, and Trains. Since 2003, the company has accumulated production, sales, and distribution data. As the company enters the second half of 2005, leadership requires production forecasts, cost analysis, and strategic profit projections to guide operations through December 2005.

Summary of Recommendations

Recommendation	Focus Area	Expected Impact
1. Halt Train Production	[Factory Production]	Omit Less Profiting Items
2. Floor Sale Prices	[Future Profits]	Increase Profits
3. Recommended Productions	[Production Volume]	Stable Production
4. Email Marketing	[Customer Engagement]	Customer Interaction and Sales

Resource Analysis

This section summarizes the findings from the analysis project. The analysis used a weighted least squares regression model to determine the cost per unit (CPU) for each product line. The model was also used to understand factory level structures.

Regression Model Overview

A weighted regression mode was fit to weekly production data across three factories influenced by outliers. The model Achieved an R^2 of 0.928, indicating that 92.8% of the variance in weekly production cost is explained by production volumes and factory assignment.

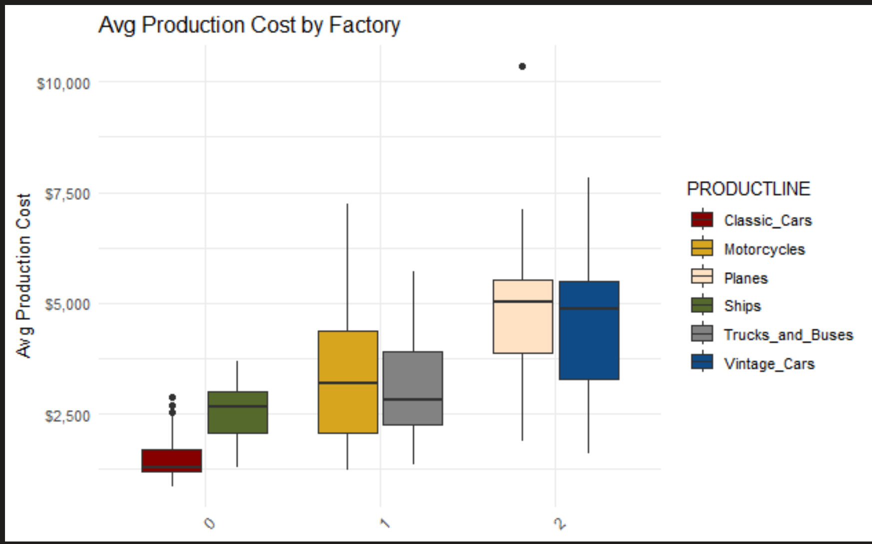
Metric	Value
R^2	0.9282
Adjusted R^2	0.9266
Residual Std. Error	226.4
Observations	355 (outliers excluded)

Cost Per Unit (CPU) by Product Line

The table below shows the regression derived CPU for each product line. These values represent the marginal cost of producing one additional unit of each product.

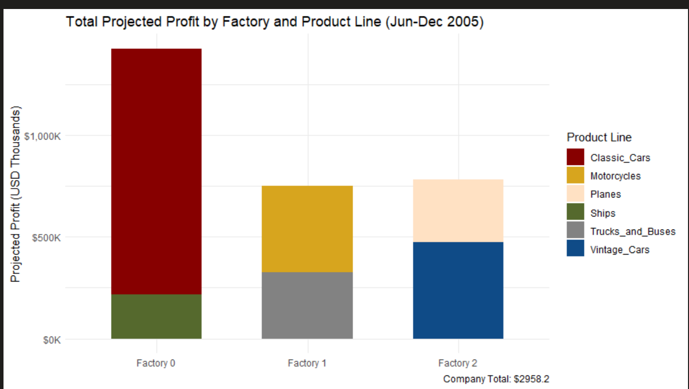
Product Line	**CPU (Regression)**	**Interpretation**
Trucks & Buses	$91.04	Highest cost per unit
Classic Cars	$83.73	High cost, low volume
Planes	$83.10	High cost, moderate volume
Motorcycles	$75.09	Mid-range cost
Vintage Cars	$66.67	Moderate cost
Ships	$15.91	Lowest CPU, highest volume

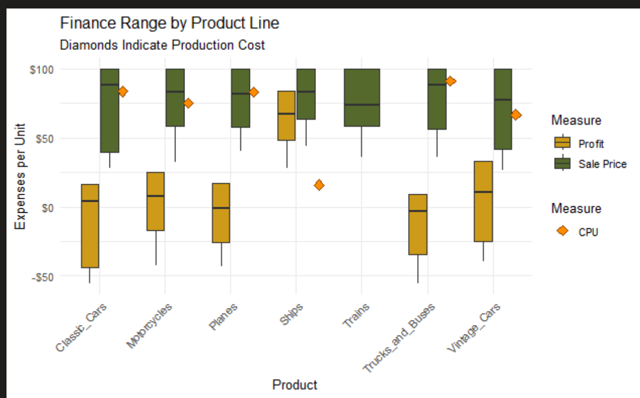
Factory Performance



The regression model identified a base operating cost (intercept) for each factory, representing fixed overhead such as labor and electricity. Factory 0 carries a base cost of $448.03 per week and serves as the model baseline. Factory 1 operates at a reduced base cost of $256.69, approximately $191 less than Factory 0. Although this difference carries a p-value of 0.48, indicating limited statistical significance. Factory 2 has the highest base cost at $546.77, roughly $98 more than Factory 0, and also shows the highest average monthly production costs overall (exceeding $7,000 in some periods). Factory 2 produces Planes and Vintage Cars, two lines with notably different CPUs, and the high cost of Plane production appears to drive up the blended cost for that facility.

Key Takeaways





Classic Cars are the most profitable product line despite relatively high production costs ($83.73/unit) and low production volumes. Ships have the lowest CPU ($15.91) and the highest production counts, yet rank among the least profitable due to low selling prices. An inverse relationship was observed between Classic Cars (low production, high sales) and Ships (high production, lower sales), suggesting a significant back stock dynamic that warrants internal review. Instances of items being sold below their regression CPU were also identified across multiple product lines, representing a key risk to margin health. Train production data was absent entirely, and production should be halted until complete records are available.

Recommendations

Recommendation 1: Halt Train Production & Align Production to Seasonality

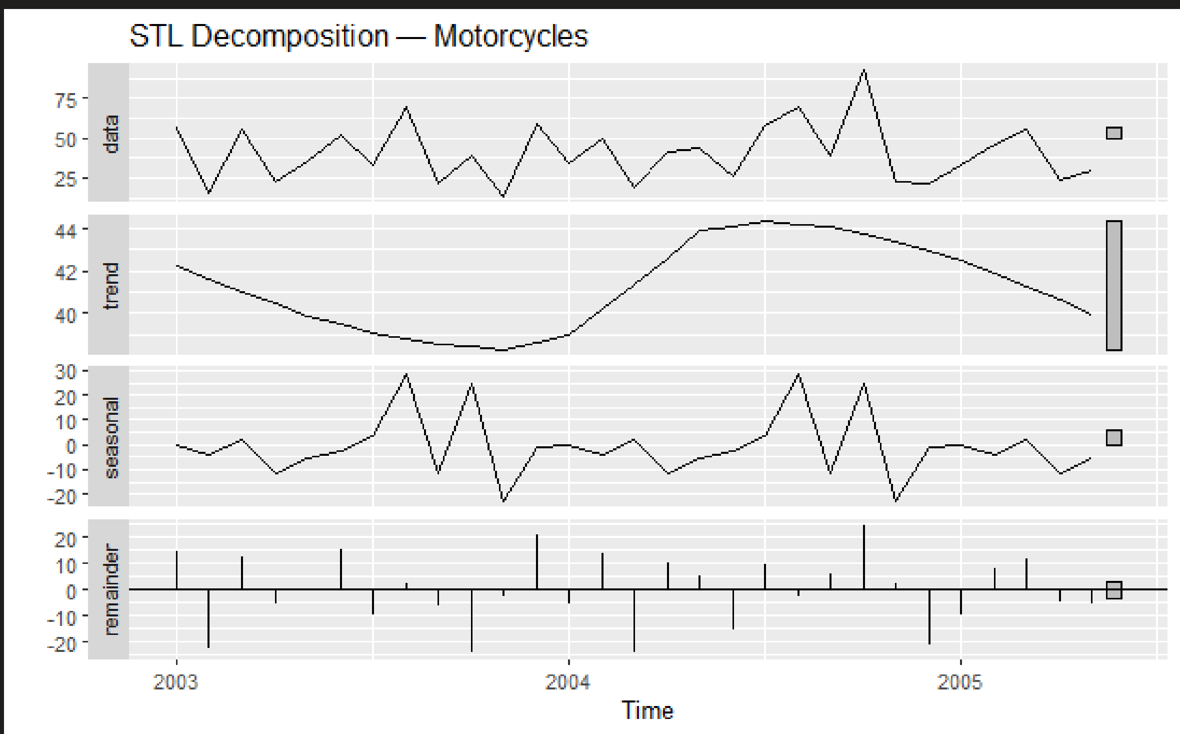
Background

Train production data is entirely absent from the current dataset, making it impossible to plan costs, volumes, or forecasts for this line. Continuing to allocate factory time and resources to Trains without data creates invisible risk. Additionally, all active product lines show clear seasonal production patterns with peaks in August and November–December. Aligning production schedules to these cycles will ensure adequate inventory during high-demand periods

Supporting Data

STL time series decomposition of all active product lines confirmed consistent, repeating seasonal patterns. Production forecasts (June–December 2005) were generated using STL models and reveal expected holiday surges for Motorcycles, Planes, Ships, and Vintage Cars.

See: STL Decomposition plots and Production Forecast charts for more visuals (Project_3_Technical_Report.Rmd)



Action Steps

1. [Specific action step 1]
2. [Specific action step 2]
3. [Specific action step 3]

Expected Impact

[Describe the expected outcome if this recommendation is implemented. Tie back to data where possible.]

Recommendation 2: [Your Recommendation Title Here]

Background

[Describe the problem or opportunity this recommendation addresses. Keep it non-technical and focused on business impact.]

Supporting Data

[Reference the specific analysis that supports this recommendation. Include a chart or table here.]

Insert visualization: [Chart title and source file]

Metric	Finding
[Metric 1]	[Value / insight]
[Metric 2]	[Value / insight]
[Metric 3]	[Value / insight]

Action Steps

1. [Specific action step 1]

2. [Specific action step 2]
3. [Specific action step 3]

Expected Impact

[Describe the expected outcome if this recommendation is implemented. Tie back to data where possible.]

Recommendation 3: [Your Recommendation Title Here]

Background

[Describe the problem or opportunity this recommendation addresses. Keep it non-technical and focused on business impact.]

Supporting Data

[Reference the specific analysis that supports this recommendation. Include a chart or table here.]

Insert visualization: [Chart title and source file]

Metric	Finding
[Metric 1]	[Value / insight]
[Metric 2]	[Value / insight]
[Metric 3]	[Value / insight]

Action Steps

1. [Specific action step 1]
2. [Specific action step 2]
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Expected Impact

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